RAI-818: INTELLIGENT MOBILE ROBOTICS

Kalman Filter – II

- **Developing a Kalman Filter**
- **Examples**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations

H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:

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- **Kalman Filter**

Kalman Filter

$$P(H \mid D) = \frac{P(H) \times P(D \mid H)}{P(D)}$$

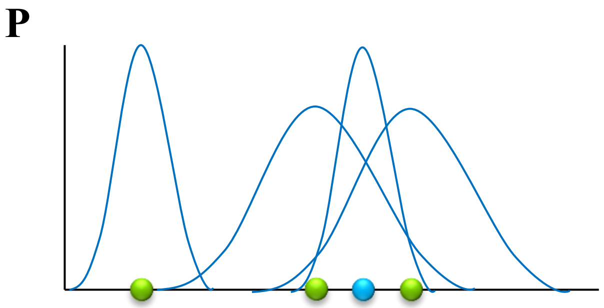
- **$P(H_t \mid H_{t-1}, D_t)$**
- However, we assumed that the state is constant and is not moving.
- What if we want to model something more complex?
- Say, we have a robot which just goes around moving people
- In addition to moving people the robot needs to know where he is?
- Let's say it is fitted with a GPS; however, the GPS has a lot of errors.

Defining futures

- Kalman Filter

Kalman Filter

- P is the prob of where the robot thinks it is located.
- Now the robot moves to the new point.
- The robot could figure this out from the command he sends out and the sped of the wheels. However, this will not be accurate since there is a lot of noise
- Now he gets data from GPS which thinks that it is somewhere else
- We can combine these to build an estimate of where the robot actually is



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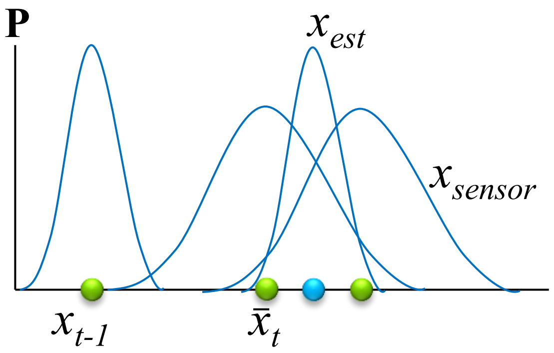
- Kalman Filter

Kalman Filter

- This is what the Kalman Filter is going to be estimating
  $P\,(H_t|H_{t-1}, A_t, D_t)$

- The additional variable A_t, or the command variable, which is affecting our estimate, allows us to have a moving state

- We need to get a good value of x_{est} in a computationally efficient manner by using a Kalman filter



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- **Kalman Filter**

Kalman Filter

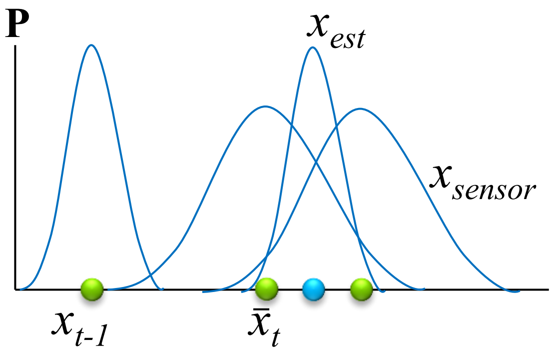
- Step I – State Prediction: How to calculate $\bar{x}_t$?

- Kalman Filter is a linear system of equations and (predicted state is a linear function of the prior state and the command

$$\bar{x}_t = Ax_{t-1} + Bu_t + \varepsilon_t$$

- The type of linear function depends on the physics and rules defined in ***A*** and ***B***

- We also have a Gaussian error term or noise



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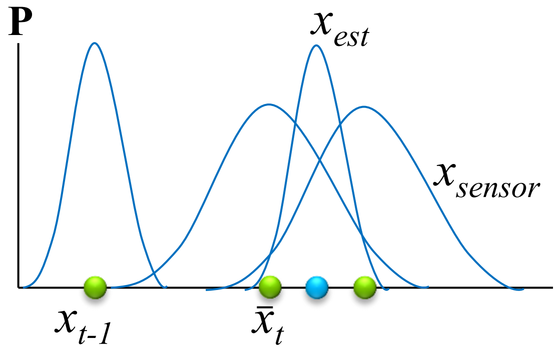
• Kalman Filter

Kalman Filter

- Step II – State Correction: What kind of information can be received from the sensor (GPS)? $\bar{z}_t = C\bar{x}_t + \varepsilon_z$
- Sensor noise is also Gaussian
- From this we get

$$x_{est} = \bar{x}_t + \underbrace{K(z_t - \bar{z}_t)}_{correction\ term}$$

- K is the Kalman Gain
- We have a predicted estimate and a sensor prediction
- If sensor prediction is accurate and equals the measurement than I can just go with my prediction.
- If not and my sensor estimate is different from the measurement than the predicted position is wrong and needs to be corrected by the Kalman Gain



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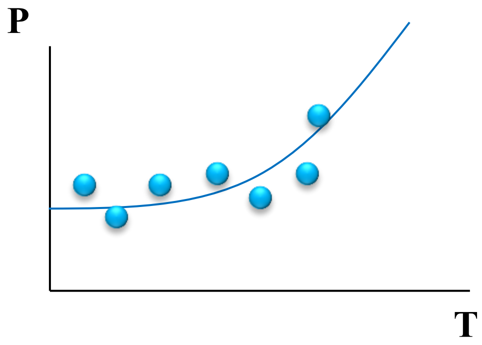
Kalman Filter - Example

- Let's return to our researcher and last time that we saw him he had accurately determined the location of the frog from the sounds.
- However, as soon as he tries to attack and capture the frog, he finds that the frog is super fast and he jumps away.
- The researcher immediately jumps after the frog, however, since it is night, the researcher cannot see very well
- Now he wants to find the frog and quickly but there is a large amount of noise regarding the location of the frog
- Now since he has an iPad and is intelligent, he decides to use a Kalman filter to track the frog



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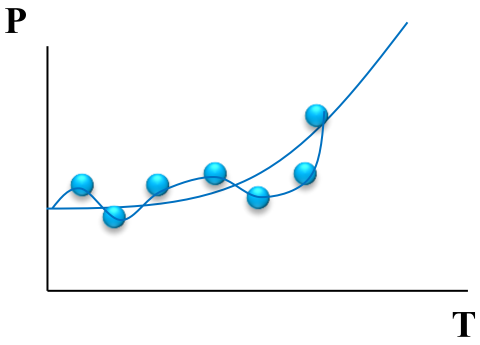
Kalman Filter

- Kalman Filter
- **Example**

Kalman Filter - Example

- Now instead of following a weird trajectory the researcher would like to follow the curve

- The frog is changing its position which the researcher will model with natural physics and motion (velocity, acceleration, etc.) to get the best estimate of the frog location.



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Kalman Filter - Example

- The first thing that we are going to model is that state prediction

$$\bar{x}_t = Ax_{t-1} + Bu_t + \varepsilon_x$$

- Let's model the state as position and velocity

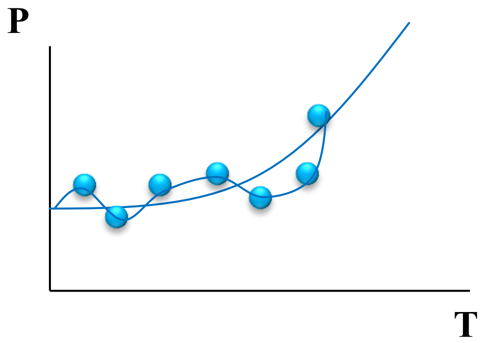
$$x_t = \begin{bmatrix} p \\ v \end{bmatrix}$$

- The standard eqns for the model will be

$$p_t = p_{t-1} + v_{t-1}t + \frac{1}{2}a_t t^2$$

$$v_t = v_{t-1} + a_t t$$

- We are going to put this into our model



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Kalman Filter

- Kalman Filter
- Example

Kalman Filter - Example

- Our state prediction becomes

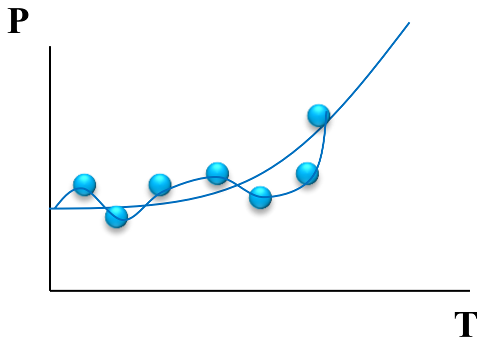
$$\bar{x}_t = \begin{bmatrix} p_t \\ v_t \end{bmatrix} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p_{t-1} \\ v_{t-1} \end{bmatrix} + \begin{bmatrix} t^2/2 \\ t \end{bmatrix} a_t + \varepsilon_x$$

- Now we will take the measurement prediction

$$\bar{z}_t = C\bar{x}_t + \varepsilon_z$$

- Since we are only measuring the position, this will be

$$p_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} p_{t-1} \\ v_{t-1} \end{bmatrix} + \varepsilon_z$$



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Kalman Filter - Example

- Lets see all the terms before we get into the nastiness of the Kalman filter

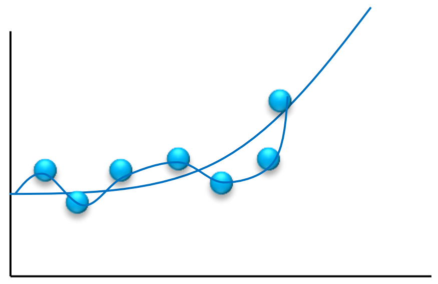
$$\bar{x}_t = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p_{t-1} \\ v_{t-1} \end{bmatrix} + \begin{bmatrix} t^2/2 \\ t \end{bmatrix} a_t + \varepsilon_x$$

$$\bar{z}_t = C\bar{x}_t + \varepsilon_z$$

- The Kalman Filter requires that the errors be written in the form of covariance matrices as follows:

$$\boldsymbol{\varepsilon_x = E_x = \begin{bmatrix} \sigma_p^2 & \sigma_p\sigma_v \\ \sigma_v\sigma_p & \sigma_v^2 \end{bmatrix}}$$

$$\boldsymbol{\varepsilon_z = E_z = \sigma^2 = 1(ft)}$$



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Kalman Filter

Kalman Filter - Equations

- Now we can combine all these things into a single nasty equation which is an explicit representation of the Kalman Filter

$$\overline{x}_t = A x_{t-1} + B u_t$$
$$\overline{\Sigma}_t = A \Sigma_{t-1} A^t + E_x$$
$$K_t = \overline{\Sigma}_x C^t (C \overline{\Sigma}_x C^t + E_z)^{-1}$$
$$x_t = \overline{x}_t + K_t (z_t - C \overline{x}_t)$$
$$\Sigma_t = (I - K_t C) E_x$$

- As the Error in measurement term gets larger in the Kalman Gain equation the value of K_t gets smaller

- This gain is multiplied in the correction term therefore, if the measurement error gets large we are going to trust it less and less hence the value of the Kalman Gain is decreased

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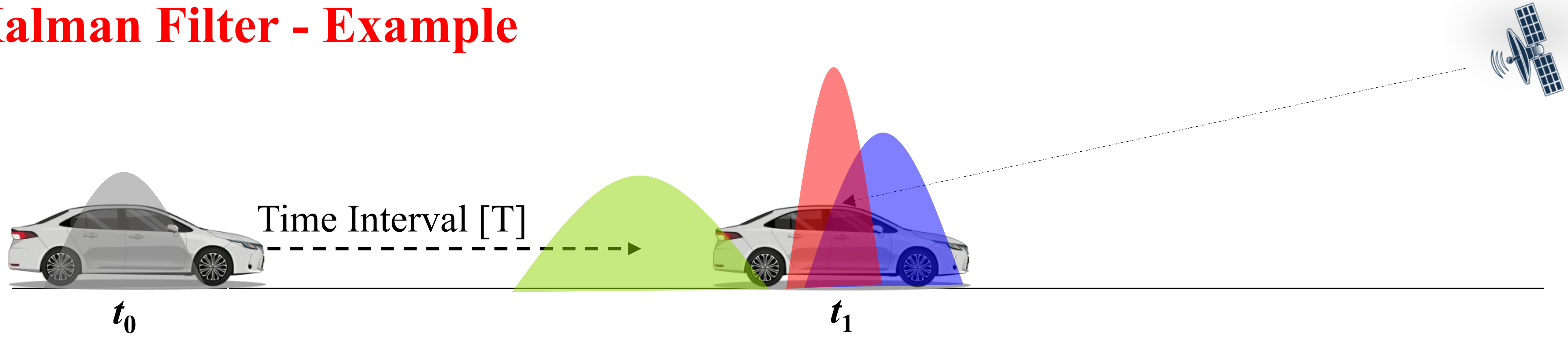
Kalman Filter

- Kalman Filter
- **Example**

Kalman Filter - Example



Prediction

$\vec{x}_{k-1,k-1}$, $P_{k-1,k-1}$ → $$\vec{x}_{k,k-1} = F_k \vec{x}_{k-1,k-1} + B_k u_k$$ $$P_{k,k-1} = F_k P_{k-1,k-1} F_k^T + Q'_k$$ → $\vec{x}_{k,k-1}$, $P_{k,k-1}$

Correction

$\vec{z}_k$, R_k → $$\vec{y} = \vec{z}_k - H_k \vec{x}_{k,k-1}$$ $$S_k = H_k P_{k,k-1} H_k^T + R_k$$ → $$K_k = P_{k,k-1} H^T S_k^{-1}$$ $$\vec{x}_{k,k} = \vec{x}_{k,k-1} + K_k \vec{y}_k$$ $$P_{k,k} = P_{k,k-1} - K_k H_k P_{k,k-1}$$ → $\vec{x}_{k,k}$, $P_{k,k}$

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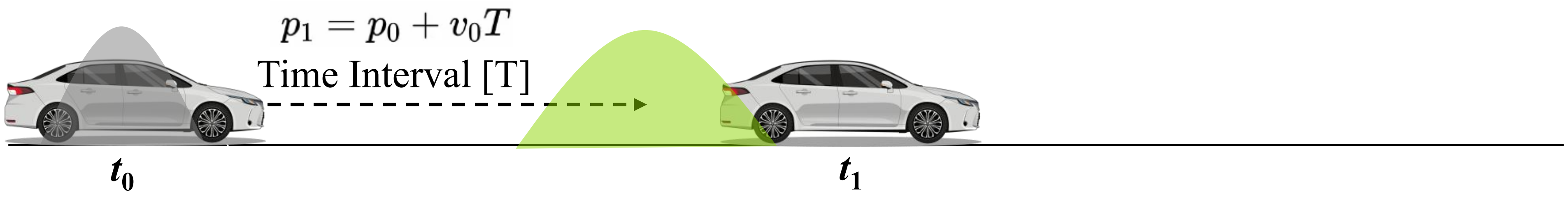
- Kalman Filter
- **Example**

Kalman Filter - Example

Prediction Step:



Initial State

$$\vec{x}_{t_0} = \begin{bmatrix} position : 0\, m \\ velocity \; : 2\, \frac{m}{s} \end{bmatrix} \Rightarrow \underset{\vec{x}_{t_1}}{\begin{bmatrix} 2 \\ 2 \end{bmatrix}} = \underset{F}{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}} \underset{\vec{x}_{t_0}}{\begin{bmatrix} 0 \\ 2 \end{bmatrix}} \Rightarrow \vec{x}^{-}_{t_1} = \begin{bmatrix} 2\, m \\ 2\, \frac{m}{s} \end{bmatrix}$$

Initial Covariance *Process Covariance*

$$P_{t_0} = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} \Rightarrow \underset{F}{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}} \underset{P_{t_0}}{\begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}} \underset{F^T}{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}} + \underset{Q}{\begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}} \Rightarrow P^{-}_{t_1} = \begin{bmatrix} 0.333 & 0.15 \\ 0.15 & 0.2 \end{bmatrix}$$

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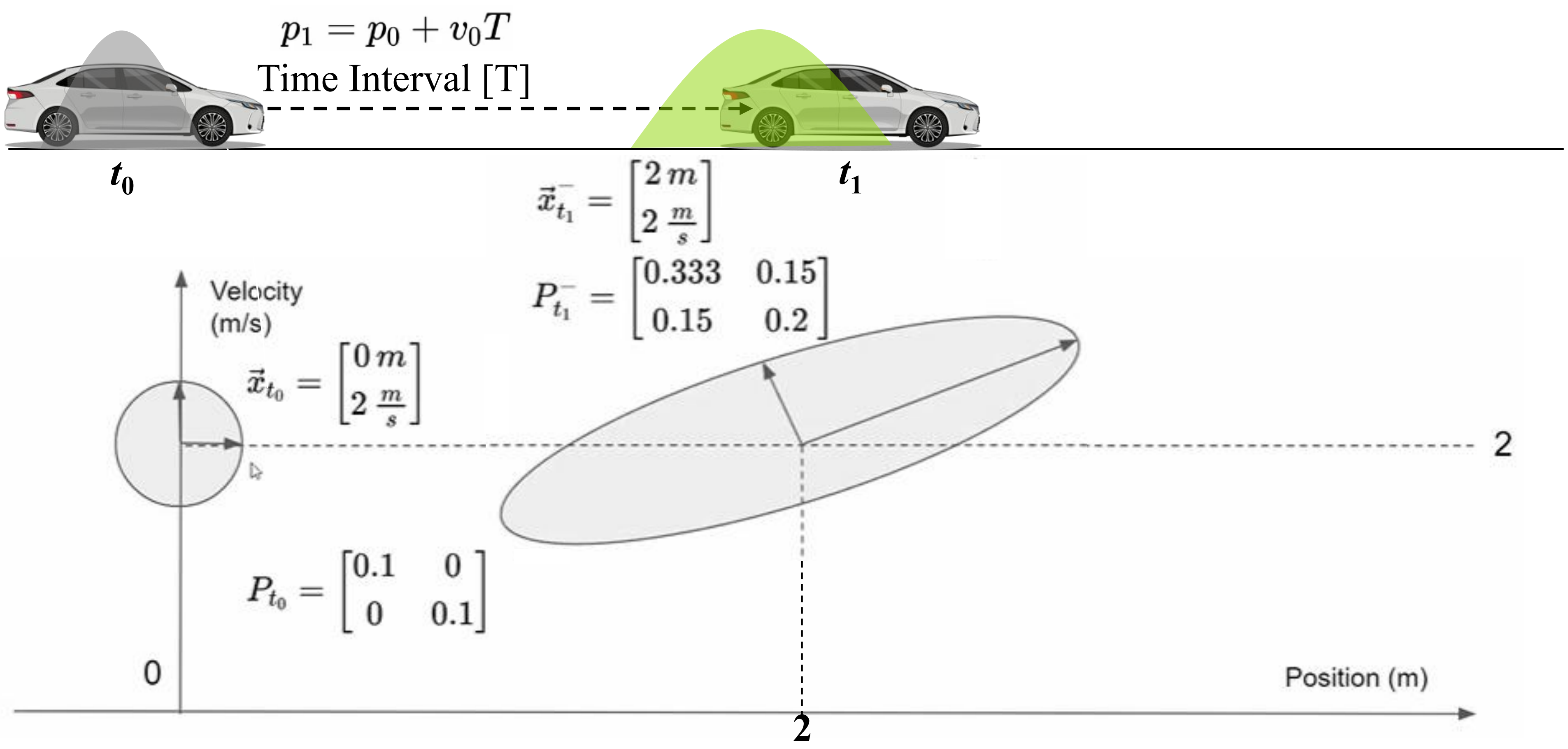
- **Example**

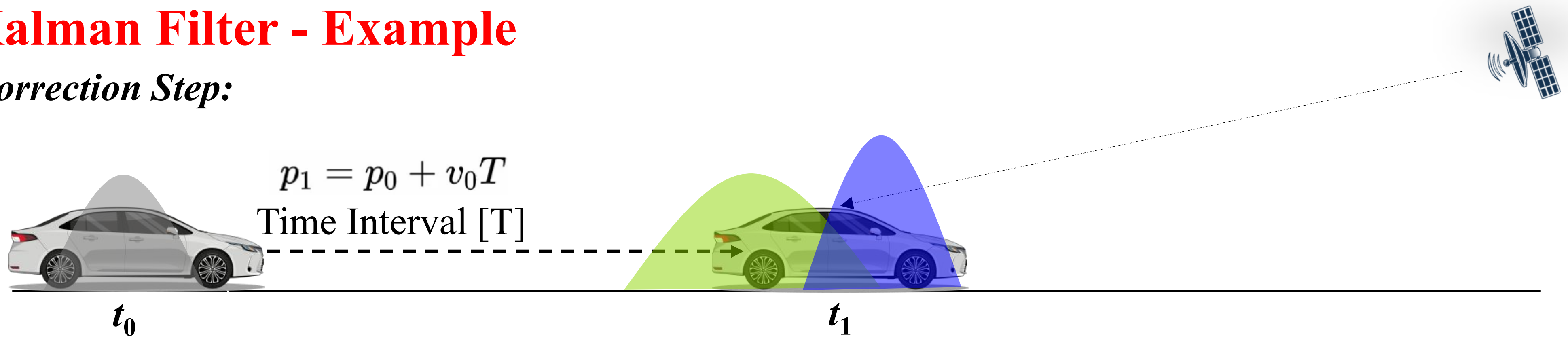
Kalman Filter - Example

Prediction Step:



Kalman Filter - Example

Correction Step:



Measurement Model

Predicted State $\vec{x}_{t_1}^{-} = \begin{bmatrix} 2\,m \\ 2\,\frac{m}{s} \end{bmatrix} \Rightarrow \overset{H}{\begin{bmatrix} 1 & 0 \end{bmatrix}} \overset{\vec{x}_{t_1}^{-}}{\begin{bmatrix} 2 \\ 2 \end{bmatrix}} \Rightarrow \vec{z}_{t_1}^{-} = [2]$ *Predicted Measurement*

$$\Downarrow$$

$$\boxed{\begin{aligned} \vec{y} &= \vec{z}_{t_1} - \vec{z}_{t_1}^{-} \\ &= 2.25 - 2 \\ &= 0.25 \end{aligned}}$$

$\vec{z}_{t_1} = [position : 2.25\,m]$

$R = [0.01]$

Actual Measurement

$$\boxed{\begin{aligned} S_{t_1} &= HP_{t_1}^{-}H^{T} + R \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0.333 & 0.15 \\ 0.15 & 0.2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + [0.01] \\ &= [0.343] \end{aligned}}$$

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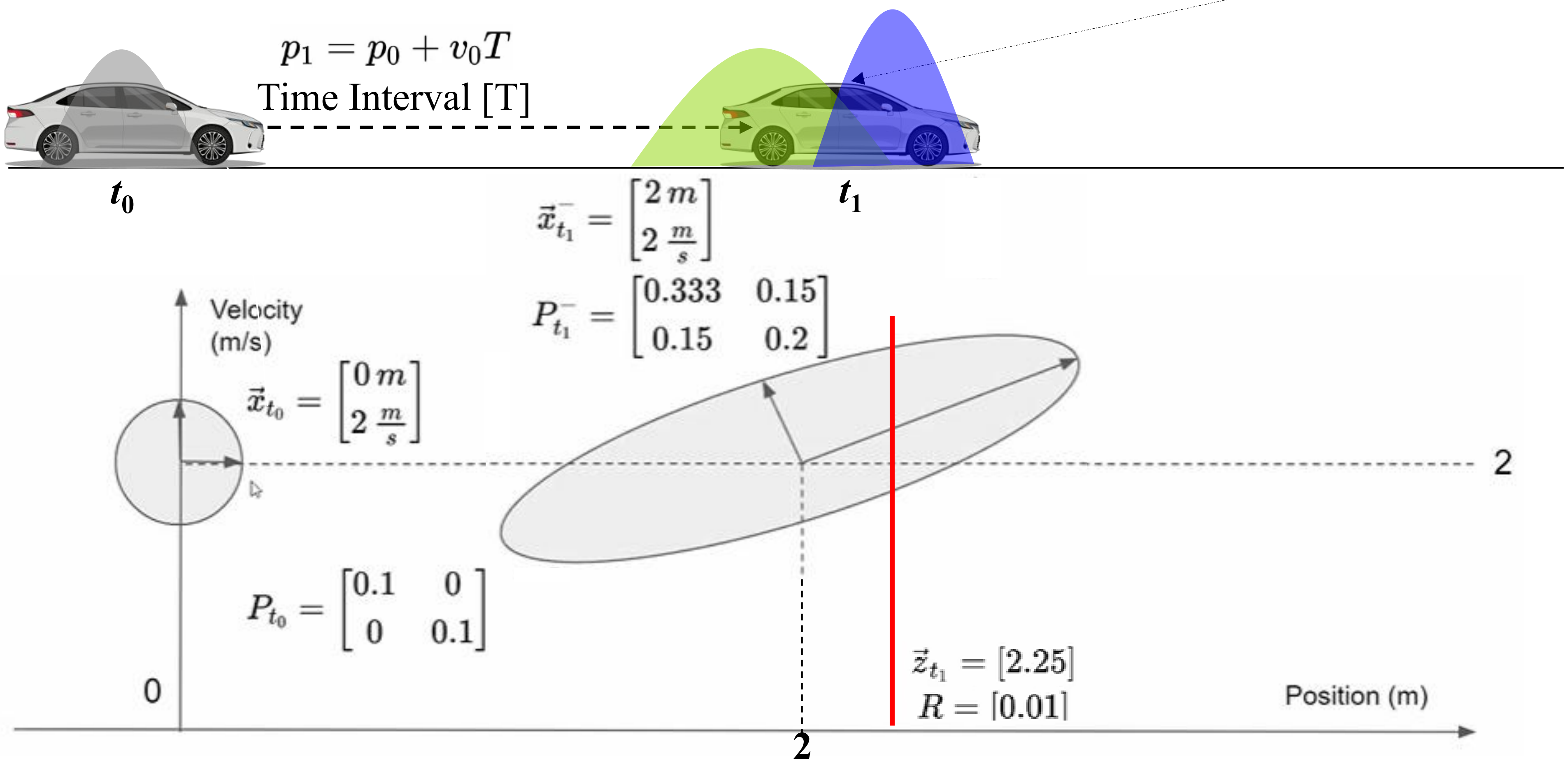
Kalman Filter - Example

Prediction Step:



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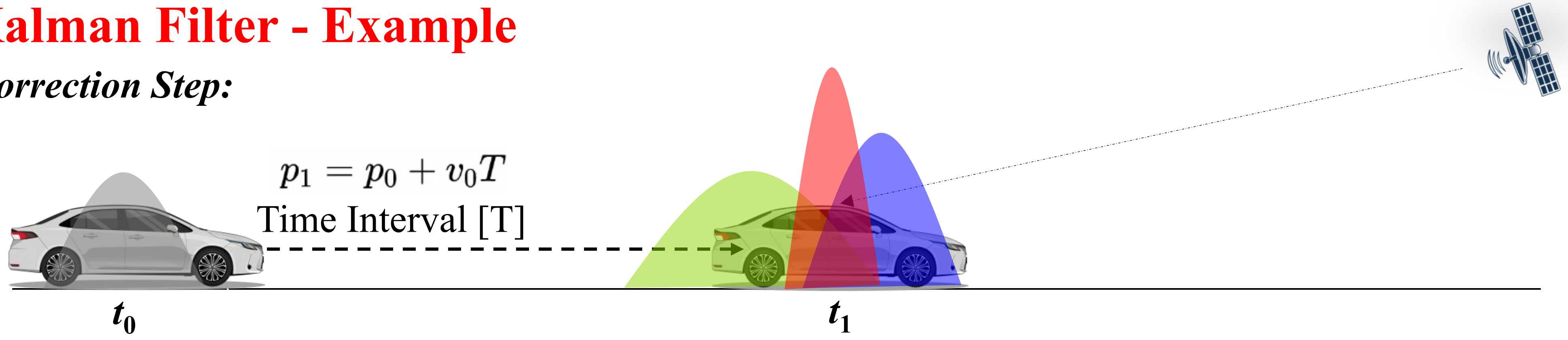
Kalman Filter - Example

Correction Step:



Kalman Gain $K_k = P_{k,k-1} H^T S_k^{-1} = \begin{bmatrix} 0.97 \\ 0.436 \end{bmatrix}$

Corrected State $\vec{x}_{k,k} = \vec{x}_{k,k-1} + K_k \vec{y}_k = \begin{bmatrix} 2.2427 \\ 2.109 \end{bmatrix}$

Corrected Covariance $P_{k,k} = P_{k,k-1} - K_k H_k P_{k,k-1} = \begin{bmatrix} 0.00970874 & 0.00436893 \\ 0.00436893 & 0.13446602 \end{bmatrix}$

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- **Example**

Kalman Filter - Example

Prediction Step: